

ARTIFICIAL INTELLIGENCE ASSIGNMENT

Topic:

***Smart Thermostat / AC Agent:
Agent Formulation and PEAS Evaluation***

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SmartCool AI

Smart Thermostat / AC Agent
Agent Formulation and PEAS Evaluation

1. Introduction

SmartCool AI is a smart air-conditioning agent that controls the temperature of a room by itself. It keeps checking how hot or humid the room is and whether anyone is inside. Based on this information, it decides when to start the AC, when to stop it and when to save power.

The aim of this agent is to give the user comfort without any manual switching and to avoid wasting electricity.

2. Agent Name and Goal

- **Name:** SmartCool AI
- **Type:** Simple reflex agent (it works on condition-action rules)
- **Goal:** Keep the room at the temperature chosen by the user while using as little electricity as possible.

3. How the Agent Works

The agent works in a repeating loop of four steps:

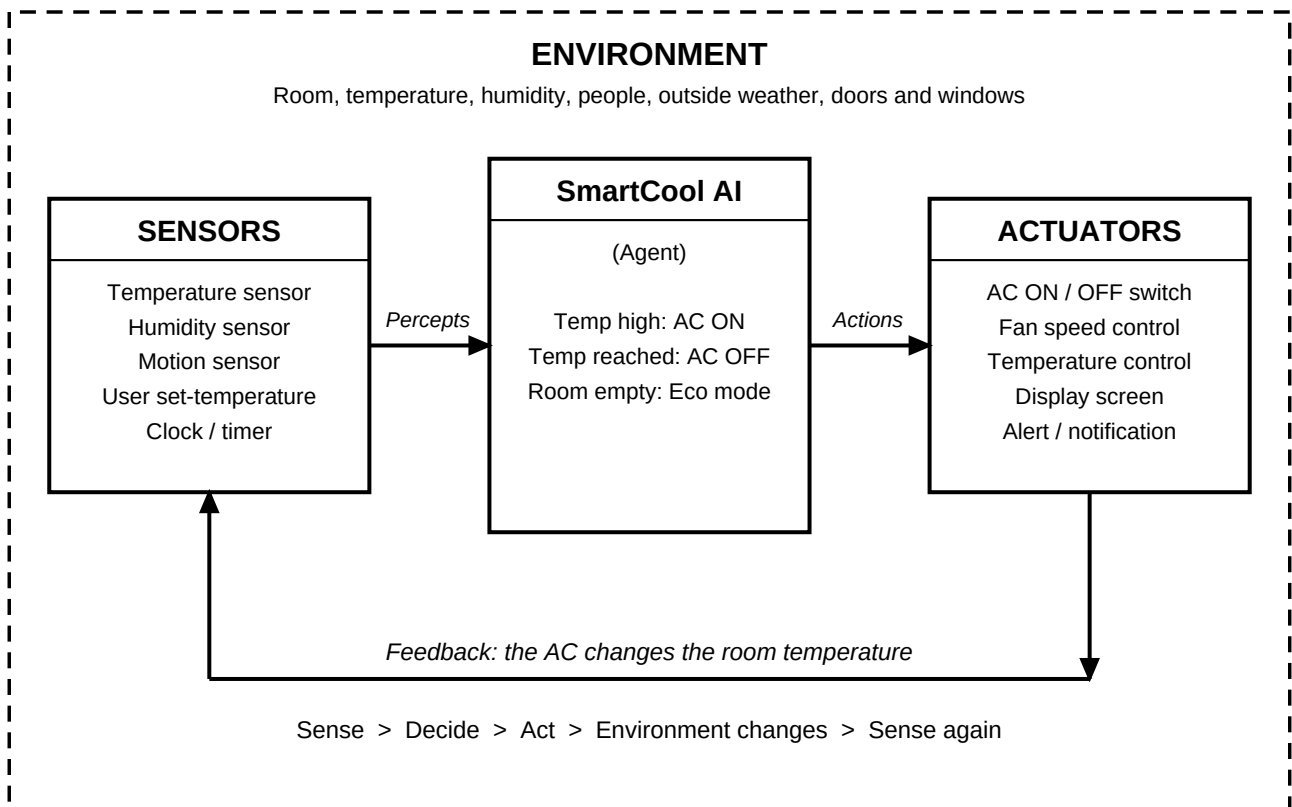
Step	What happens
1. Sense	Sensors measure room temperature, humidity, motion and time. The user's set-temperature is also taken as input.
2. Decide	The agent compares the current temperature with the set-temperature and checks its rules.
3. Act	The agent switches the AC on or off, changes fan speed, updates the display or sends an alert.
4. Feedback	The room temperature changes because of the AC, and the sensors read the new value for the next loop.

4. Agent Formulation

Component	Description
Agent	SmartCool AI (automatic AC controller)
Environment	Room, air temperature, humidity, people, outside weather, doors/windows and the AC unit
Sensors	Temperature sensor, humidity sensor, motion sensor, clock, user's set-temperature
Actuators	AC on/off switch, fan speed control, temperature control, display screen, alert buzzer
Percepts	Current temperature, humidity level, room occupied or empty, time, target temperature
Actions	Turn AC on, turn AC off, change fan speed, switch to eco mode, show alert

5. Graphical Representation

The diagram shows how SmartCool AI is connected to its environment through sensors and actuators.



Rules followed by the agent:

If (percept)	Then (action)
Temperature is 1 degree or more above the set value and room is occupied	Turn AC on
Temperature is 1 degree or more below the set value	Turn AC off
Room has been empty for 15 minutes	Eco mode / AC off
Temperature is above 40 degrees C	Maximum fan speed and show alert

6. Environment Types

Property	Type	Reason
Observability	Partially observable	A sensor reads only one spot, so the agent cannot know every corner of the room or when people will come in.
Determinism	Stochastic	Open doors, sunlight and outside weather change the room temperature in unpredictable ways.
Episodic / Sequential	Sequential	Every AC action changes the temperature and affects the next decision.
Static / Dynamic	Dynamic	Temperature and occupancy keep changing while the agent is working.
Discrete / Continuous	Continuous	Temperature and humidity are continuous values (AC on/off is discrete).
Single / Multi-agent	Single agent	Only SmartCool AI makes decisions. People and weather are part of the environment.

7. PEAS Framework

PEAS (Performance, Environment, Actuators, Sensors) describes the task environment of an agent.

Element	SmartCool AI
Performance	Room stays near the desired temperature; low electricity use; quick response; user comfort; no unnecessary on/off switching
Environment	Room, air temperature, humidity, people, outside weather, doors/windows, AC unit
Actuators	AC on/off switch, fan speed control, temperature control, display screen, alert buzzer
Sensors	Temperature sensor, humidity sensor, motion sensor, clock, user's set-temperature input

8. Example Working Cycle

- The user sets the temperature to 24 degrees C.
- The sensor reads 30 degrees C and the motion sensor finds a person in the room.
- The agent checks its rules: the room is hotter than required, so cooling is needed.
- The agent turns the AC on and selects a suitable fan speed.
- The room cools down and the sensors keep sending new readings.
- At about 24 degrees C the agent turns the AC off. If the room becomes empty, it goes into eco mode.

9. Conclusion

SmartCool AI is a good example of an intelligent agent. It senses its environment, decides using simple rules and acts through actuators to reach its goal. The PEAS description clearly shows how its success is measured, where it works, what it can do and what it can sense.